

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented RTL for high-speed **Ethernet MAC, PCS, and FEC IP cores**, supporting multi-rate, multi-port architectures up to 1.6Tb.
- Collaborated daily with a co-located team of design, verification, and back-end engineers to advance IP core development cycles.
- Learned and applied digital design best practices, including **synthesis** and **timing closure** methodologies for configurable IP delivery.
- Developed Python scripts to automate data validation and ETL processes, reducing manual data handling efforts by approximately **30%**.
- Wrote and optimized **SQL queries** across PostgreSQL and Oracle databases for data analysis and reporting tasks.
- Contributed to feature development and bug fixes for Java Spring Boot applications, focusing on **REST API enhancements** and service stability.

SKILLS

Digital Design – RTL, Verilog, SystemVerilog, Synthesis, Timing Closure, Clocking, State Machines, High-Speed Networking, Ethernet Standards

Hardware Description Languages – Verilog, SystemVerilog

Programming Languages – Python, Java, SQL

Databases – PostgreSQL, Oracle, MySQL

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Development Tools & Methodologies – Git, Jenkins, Jira, Confluence, Agile, Scrum

Analytics & BI – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy, scikit-learn

PROJECTS

High-Speed Serial Link IP Development

- Designed and simulated RTL for a **400Gb Ethernet PCS layer**, focusing on forward error correction (FEC) and link training sequences.
- Integrated custom Verilog modules into an existing IP core framework, ensuring compatibility with simulation and synthesis toolchains.
- Developed unit tests in SystemVerilog to verify functional correctness of critical blocks, achieving **95% code coverage** on key modules.

Automated Data Ingestion and Reporting Pipeline

- Engineered a Python-based ETL pipeline to ingest data from multiple sources into a central PostgreSQL data warehouse for analytical reporting.
- Implemented data validation checks and error handling within the pipeline, reducing data quality issues in downstream reports by **~20%**.
- Created automated reports using Power BI, visualizing key performance indicators for business stakeholders and enabling faster decision-making.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)